

Power and resource tradeoffs

Project: Reproducible AI Research Study

1. Your mission

Your learning target

I can explain the mechanism and terminology of power and resource tradeoffs.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pilot experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Sensitivity analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

This week you will practise:

Explain the mechanism and terminology of power and resource tradeoffs

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Pilot experiments. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

My starting explanation

This week's end goal

Explain power and resource tradeoffs using the records and boundaries of this fictional investigation.

2. Notice the evidence

Power concerns detecting a declared effect under a specified model and test. Planning depends on effect size, variability, unit independence and budget.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing. These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

List the given facts and one fact that is not supplied.

3. See how it works

Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Power and resource tradeoffs — evidence trace

Input and scope · Check or calculation · Conclusion and limit

Input and scope: For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits.

Check or calculation: Use supplied simulation results rather than an unsupported universal sample-size rule. Estimated detection frequency is detections/simulated alternative trials; false positives require null trials separately.

Conclusion and limit: Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same inputs. If a required input is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

Compare a familiar case with a changed case

For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change: Change the assumed effect or variability but keep the sample sizes.

Prediction: The detection fractions need new simulation; the resource costs stay 40 and 80.

Reason: Power depends on the data-generating assumptions, while this stated unit-cost calculation does not.

5. Words that unlock the idea

Word · Meaning

Word: Power

Meaning: The test's detection probability under a specified alternative

Word: Simulation trial

Meaning: A generated dataset under a declared model

Word: Planning effect

Meaning: The effect size used for design calculations

Use two terms to explain a step in this week's example.

6. Follow a worked example

For a fixed alternative 100 simulations detect 62 times at $n=20$ and 83 at $n=40$. Per-unit cost 3 credits.

Estimated detection 62% versus 83%; cost 60 versus 120 credits per study. These simulation frequencies depend on the supplied model; null error is unmeasured.

Compare the intermediate evidence with the final statement. Explain why each number or decision follows from the stated inputs. The conclusion belongs to this scenario and procedure; changing the inputs, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Power and resource tradeoffs

A simulated detection-and-resource trade-off. A fictional simulation repeats a study under the same assumed alternative effect, using two independent-unit sample sizes.

The facts we will use

There are 100 simulation trials per design. At $n=20$, 60 detect the specified effect; at $n=40$, 80 detect it. Each independent unit costs 2 resource units. Same test and effect assumptions are used.

Stipulated simulation and resource records

	Independent units	Detections	Trials	Cost units
Smaller design	20	60	100	40
Larger design	40	80	100	80

Read a cell using both its row and column labels.

Power and resource tradeoffs

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Estimate simulated detection

Observed detection fractions are $60/100=60\%$ and $80/100=80\%$.



2. Calculate design cost

$n=20$ costs 40 resource units; $n=40$ costs 80 units.



3. Compare the trade-off

Doubling cost adds 20 percentage points of simulated detection under these assumptions.

CHANGE ONE THING

Change the assumed effect or variability but keep the sample sizes.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Estimate simulated detection

What happens and why: Observed detection fractions are $60/100=60\%$ and $80/100=80\%$.

Step: 2. Calculate design cost

What happens and why: $n=20$ costs 40 resource units; $n=40$ costs 80 units.

Step: 3. Compare the trade-off

What happens and why: Doubling cost adds 20 percentage points of simulated detection under these assumptions.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Power is the probability of detecting a specified effect under a specified model and decision rule. Repeated simulation can estimate it; the estimate depends on effect size, variability, design and test assumptions, not sample size alone.

What this example does not establish

Only 100 trials per design produce noisy estimates. They are fictional stipulated results, not executed simulation or evidence about pupils.

7. Find and repair a mistake

Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself. The first may require a correction; the second requires an explicitly revised method and new evidence. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

8. Practise together

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Power and resource tradeoffs”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Power and resource tradeoffs. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate both simulated detection fractions and resource costs. What additional detection is observed for the extra forty units, and does that establish power for an actual classroom study?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

Predict, investigate and preserve the evidence

For a fixed alternative100 simulations detect62 times at n20 and83 at n40. Per-unit cost3 credits. Show the intermediate steps and give the conclusion with its scope.

Record · Your evidence

Record: Given input and fixed rule

Your evidence: _____

Record: Prediction before checking

Your evidence: _____

Record: Actual result, or “not run”

Your evidence: _____

Record: First mismatch and one explanation

Your evidence: _____

Record: Boundary or missing input and expected response

Your evidence: _____

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

Explain → apply → challenge

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

A report omits this qualification: Power estimates do not repair biased sampling or establish the probability a particular hypothesis is true. Explain how omitting it could change a reader’s interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

11. Connect your project

Add a labelled power and resource tradeoffs evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of power and resource tradeoffs. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence · Record

Project evidence: Component and version

Record: _____

Project evidence: Inputs permitted at prediction time

Record: _____

Project evidence: Trace and comparison with earlier work

Record: _____

Project evidence: Failure, limit and human review

Record: _____

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

12. Show what you know

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer. For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and

the evidence that would make you revise your explanation.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

13. Study a complete case

Complete evidence record

Power and resource tradeoffs — evidence trace

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The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the input through the stated procedure to its result. Cover the result first, predict it, then explain

every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

14. Try a new case independently

Independent case: Apply a new case independently

One hundred alternative trials detect 70 at $n=30$ and 85 at $n=60$; unit cost 2. Compare.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

One hundred alternative trials detect 70 at $n=30$ and 85 at $n=60$; unit cost 2. Compare.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.